

Francisco Angulo de Lafuente

Research Architect · AI Systems · Formal Verification ·
Decentralized Networks

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PROFESSIONAL SUMMARY

Independent researcher, systems architect, and writer with 25+ years of experience spanning biotechnology, AI systems design, decentralized networks, and science communication. I work remotely by design — I am on the autism spectrum, which means I produce my best work in a focused, structured environment. This is not a limitation: it is why I have been able to sustain deep technical work across multiple decades and domains.

Current focus: P2PCLAW / MENTAT — a decentralized AI research network with Lean 4 formal verification and post-quantum cryptography, directed with an international team of doctoral researchers. Three peer-reviewed papers on arXiv (cs.NE, 2025). Active institutional engagements: Harvard University (Zitnik Lab), Open Source Initiative, MIT Technology Review.

PEER-REVIEWED PUBLICATIONS — ARXIV 2025

arXiv profile: arxiv.org/a/angulodelafuente_f_1 | ORCID: orcid.org/0009-0001-1634-7063

arXiv:2601.12032 Speaking to Silicon: Neural Communication with Bitcoin Mining ASICs Demonstrates that SHA-256 ASIC mining hardware can encode neural signal patterns, enabling ultra-low-cost physical neural hardware from repurposed mining equipment.

arXiv:2601.09557 SiliconHealth: A Complete Low-Cost Blockchain Healthcare Infrastructure for Resource-Constrained Regions Using Repurposed Bitcoin Mining ASICs Complete healthcare data infrastructure using decommissioned Bitcoin ASICs as edge computing nodes — secure patient records with offline-capable processing for under-resourced medical settings.

arXiv:2601.01916 Toward Thermodynamic Reservoir Computing: Exploring SHA-256 ASICs as Potential Physical Substrates Demonstrates that thermal and electrical dynamics of mining hardware exhibit the nonlinear memory properties required for reservoir computing — enabling a new class of physical neuromorphic devices.

KAGGLE — COMPETITIVE RESEARCH (PHYSICS-BASED AI ARCHITECTURES)

I use Kaggle competitions as a laboratory for testing novel computational architectures. In every case below, while thousands of teams fine-tuned standard models, I designed and implemented entirely new physical computing paradigms. Rankings are not the contribution — the architectures are.

NeurIPS — Ariel Data Challenge 2025 — Participant

Architecture: ARIEL Real Quantum-Optical Physics Simulation

Instead of training a regression model on labelled spectra, built a full quantum-optical simulator replicating the physical instrument response of ESA's Ariel space telescope — photon noise model, detector systematics, optical transfer functions. The approach treated the challenge as a physics problem, not a statistical one.

Grand X-Ray Slam (Divisions A & B) — Participant

Architecture: NEBULA — Physics-Based Neural Network (3.7M parameters)

While other teams fine-tuned ResNet / EfficientNet, designed NEBULA: a ray-tracing neural network modelling X-ray photon transport physically — attenuation coefficients, scatter, beam hardening — using those physical priors as inductive biases. Division B included SLAM-style spatial mapping of pathology across imaging sessions.

MABe 2025 — Social Action Recognition in Mice — Participant

Architecture: QESN — Quantum Echo State Network

Built a Quantum Echo State Network (reservoir computing + quantum-inspired dynamics) for classifying social behaviour from mouse pose sequences. Standard submissions used temporal CNNs or Transformers; QESN

treats the pose sequence as a physical dynamical system and uses the reservoir's nonlinear memory to encode behavioural state.

ARC Prize 2025 · RSNA Intracranial Aneurysm Detection · NeurIPS Code Golf 2025 · Digit Recognizer · Histopathologic Cancer Detection · and others — Participant

Architecture: Various experimental architectures (holographic associative memory, physical neural nets, ASIC-based inference)

Participated as a research laboratory — each competition an opportunity to test a novel computational architecture rather than optimise an existing one. Work from these experiments fed directly into three arXiv papers published in 2025 (cs.NE: reservoir computing with SHA-256 ASICs as physical substrates).

Kaggle profile (53 works): kaggle.com/franciscoangulo

PRIMARY RESEARCH PROJECT — P2PCLAW / MENTAT (2023 – PRESENT)

Role: Lead Architect & Principal Investigator | **Team:** 8 doctoral researchers, 5 countries

URLs: p2pclaw.com · github.com/Agnuxo1/OpenCLAW-P2P · app.p2pclaw.com

Technical Architecture

- HeytingLean: 3,325 Lean 4 files · 760K+ lines · zero sorry · zero admit
- Nucleus operator R on Heyting algebra — $R(x)=x$ formal publication gate
- AgentHALO: ML-KEM-768 + ML-DSA-65 post-quantum crypto (FIPS 203/204)
- DID sovereign identity · Nym mixnet · IPA/KZG polynomial proofs
- GUN.js + IPFS decentralized P2P mesh — no central server
- La Rueda: permanent IPFS archive · SHA-256 content hashing
- Swarm Compute: HuggingFace Spaces + Railway
- MCP server — standard Model Context Protocol integration
- 8-domain research lab: Physics, Chemistry, Biology, AI/ML, Robotics, Genomics, Mathematics, DeSci

Institutional Engagements

- Harvard University Zitnik Lab — TxAgent collaboration discussion
- Open Source Initiative — licensing review, April 2026 (Deb Bryant + Simon Phipps)
- MIT Technology Review — editorial interest, AI agents reporter
- Princeton Gen-Verse Lab (Dr. Ling Yang, OpenClaw-RL #1 HuggingFace Daily Papers Mar 2026)
- Show HN published March 2026

Research Paper

[OpenCLAW-P2P v4.0 — Architecture Paper \(2026\)](#)

SELECTED OPEN-SOURCE PROJECTS (92 PUBLIC REPOSITORIES)

CHIMERA Architecture — Physics-based, hardware-agnostic neural architecture — 43× speedup vs PyTorch, 88.7% memory reduction. Demonstrates optical computing principles on conventional silicon.

Enhanced Unified Holographic Neural Network (EUHNN) — Winner of the NVIDIA + LlamaIndex Developer Contest 2024. Combines holographic memory with quantum-inspired computing.

The Living Agent — Autonomous research AI navigating a 16×16 Lean 4-grounded knowledge grid (Karpathy autoresearch-inspired). Runs locally on Qwen3.5-9B via KoboldCPP with soul.md memory persistence.

OpenCLAW-2 Autonomous Multi-Agent Scientific Research Platform — Full autonomous research pipeline: literature review → hypothesis → experiment → paper writing. Python, multi-agent orchestration.

P2PCLAW MCP Server — Model Context Protocol server — any AI agent (Claude Desktop, Cursor, ChatGPT) connects to P2PCLAW in minutes.

ASIC-RAG-CHIMERA — Retrieval-augmented architecture using ASIC hardware as computing substrate. Direct precursor to three arXiv papers (2025).

Full repository list: github.com/Agnuxo1

ECOFA — BIOFUELS RESEARCH COMPANY (FOUNDED 2001, AGE 25)

Role: Founder & Research Director

Founded and directed Ecofa, developing the first second-generation biofuel from bacteria fed on organic waste — entirely separating biofuel production from food-crop competition. Multiple international patents. Published in Revista de Energías Renovables, 22(4). Covered in El País and national science media.

Patents: patents.google.com/?inventor=Francisco+Angulo+Lafuente

PUBLISHED WORKS — FICTION & SCIENCE COMMUNICATION

Published with Editorial Planeta, Editorial Alfaguara, and Penguin Random House, writing at the intersection of science fiction, environmental science, and technology.

La Reliquia — Editorial Planeta, 2006

Ecofa — Editorial Alfaguara, 2008

Kira and the Ice Storm — Penguin Random House, 2009

Eco-fuel-FA — Editorial Planeta, 2010

Los Mejores — Editorial Alfaguara, 2010

La leyenda de los Tarazashi — Penguin Random House, 2010

El Olfateador — Editorial Planeta, 2010

Diario de un boina verde — Editorial Alfaguara, 2011

Destino la Habana — Penguin Random House, 2011

Wikipedia: es.wikipedia.org/wiki/Francisco_Angulo_de_Lafuente

EDUCATION & TRAINING

Non-traditional path by design. I am on the autism spectrum and learn most effectively through focused independent study. All formal studies completed through distance / online institutions.

UNED (Universidad Nacional de Educación a Distancia) — Spain — Computer Science coursework

Universidad Politécnica de Madrid — Spain — Computer engineering and biotechnology

Continuing Professional Development — AI, Machine Learning, Distributed Systems, Cryptography, Formal Methods — Coursera, edX, and specialist providers (2015 – present)

TECHNICAL SKILLS

Programming & Systems

- TypeScript / JavaScript (Node.js, full-stack)
- Python (ML pipelines, scientific computing)
- Lean 4 (formal verification, theorem proving)
- GUN.js · IPFS · Docker · Railway · HuggingFace

AI & Machine Learning

- PyTorch · JAX · DeepSpeed · Optuna · Ray
- Multi-agent systems design and orchestration
- Physical computing: reservoir, holographic, neuromorphic
- LLM integration (local KoboldCPP + cloud APIs)

Security & Cryptography

- Post-quantum: ML-KEM-768 · ML-DSA-65 (FIPS 203/204)
- DID sovereign identity · Nym mixnet
- IPA/KZG polynomial commitment proofs
- Blockchain / Web3 · Solidity

Scientific Domains

- Biotechnology & biofuels (domain expert)
- Formal mathematics and constructive logic
- Physical neural architectures (ASIC substrates, ray-tracing)
- Science communication — Spanish & English

WORK ARRANGEMENT

Remote-first by design.

20+ years of productive remote work. Being on the autism spectrum means sustained deep focus is my natural working state — making remote, asynchronous, independent research an ideal fit, not a compromise. Fully available for: scheduled video conferences, written technical collaboration in English and Spanish, online seminars and technical guidance, and periodic visits to China (1–2 per year, 2–3 months total, with advance planning). The Zhejiang University Shaoxing Research Institute arrangement is a format I am willing to explore with appropriate support.

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patents.google.com/?inventor=Francisco+Angulo+Lafuente · researchgate.net/profile/Francisco-Angulo-Lafuente-3